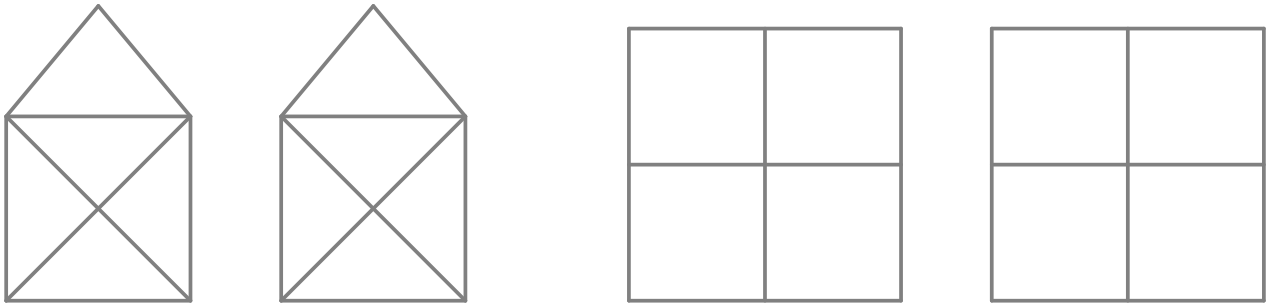
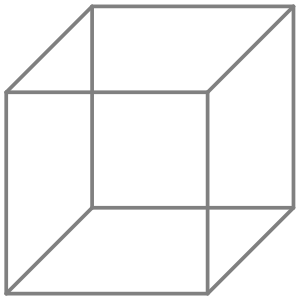
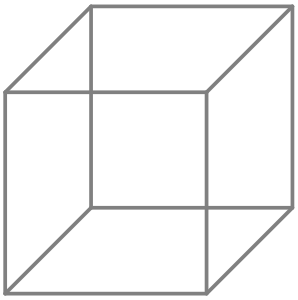
Put a counter on every bridge of a town before you walk it. A walk goes from island to island across bridges, and crossing a bridge means picking up its counter. A bridge whose counter is gone cannot be crossed again. Two islands can be joined by more than one bridge. In every town you can get from any island to any other.

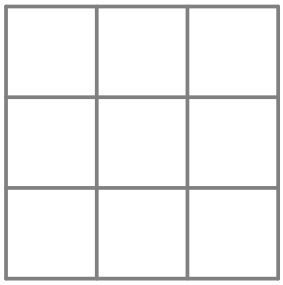
Problem 6: Can a town have exactly one island with an odd number of bridges? Can it have exactly three? Explain.

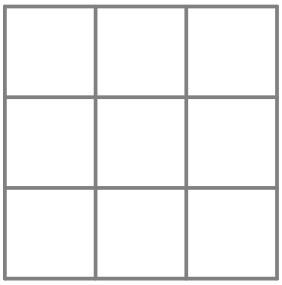
Problem 7: Trace each picture with a colored pencil, without lifting the pencil and without going over a line twice, or explain why it cannot be done. For each picture that cannot be traced this way, find the fewest strokes that draw it without going over a line twice, using a new color for each stroke.

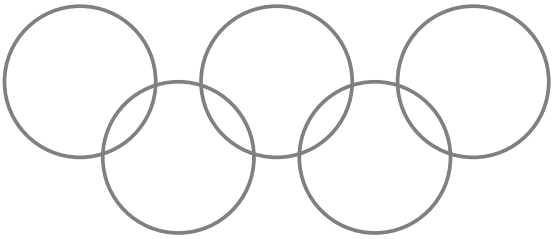


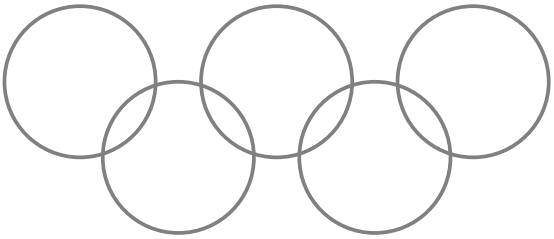


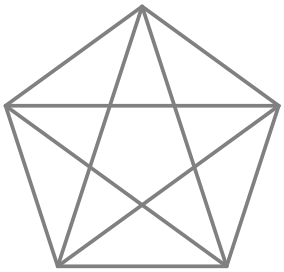


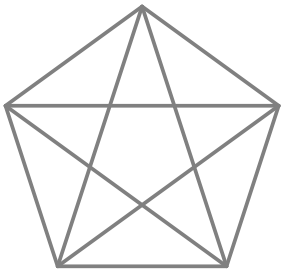




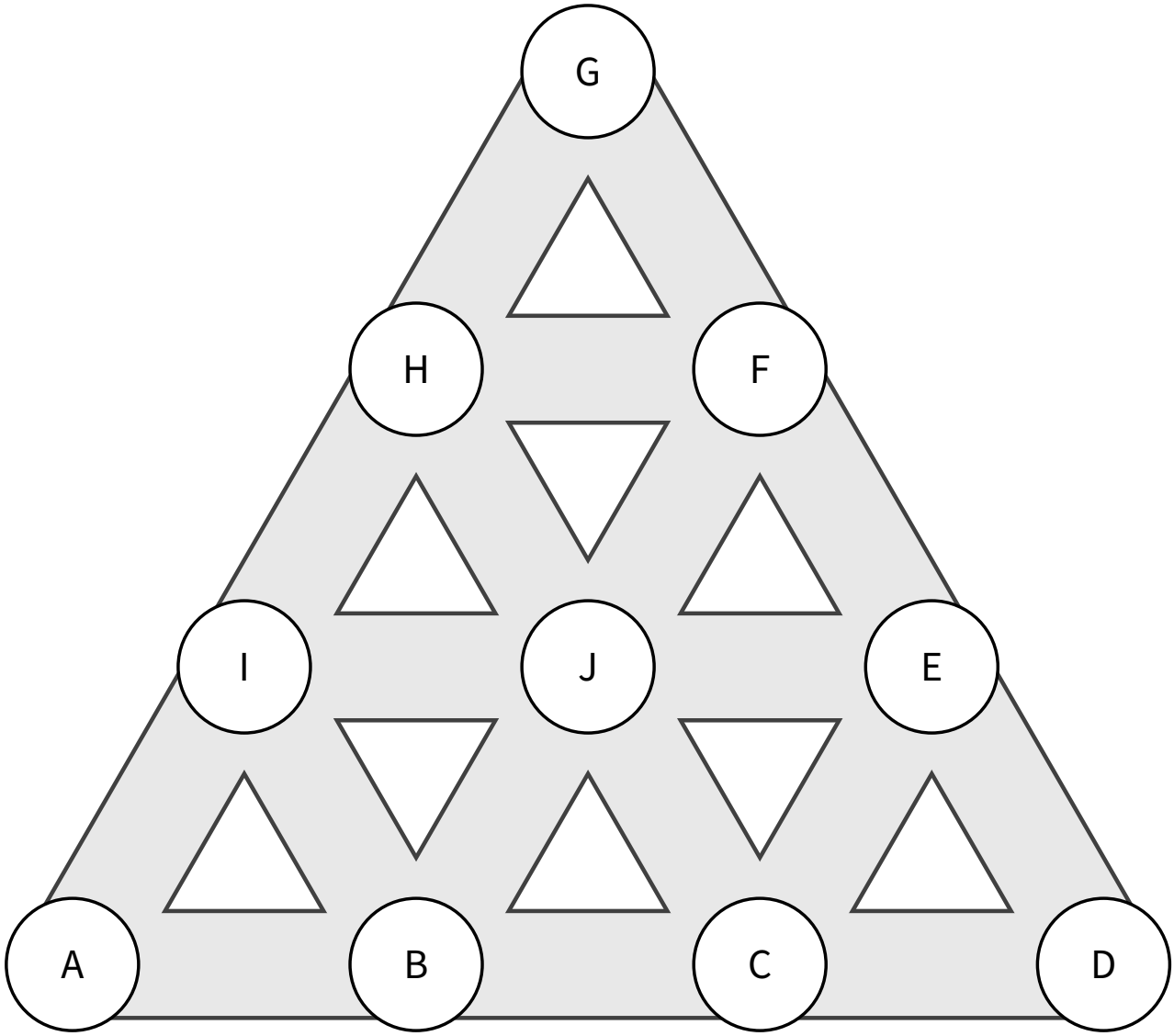






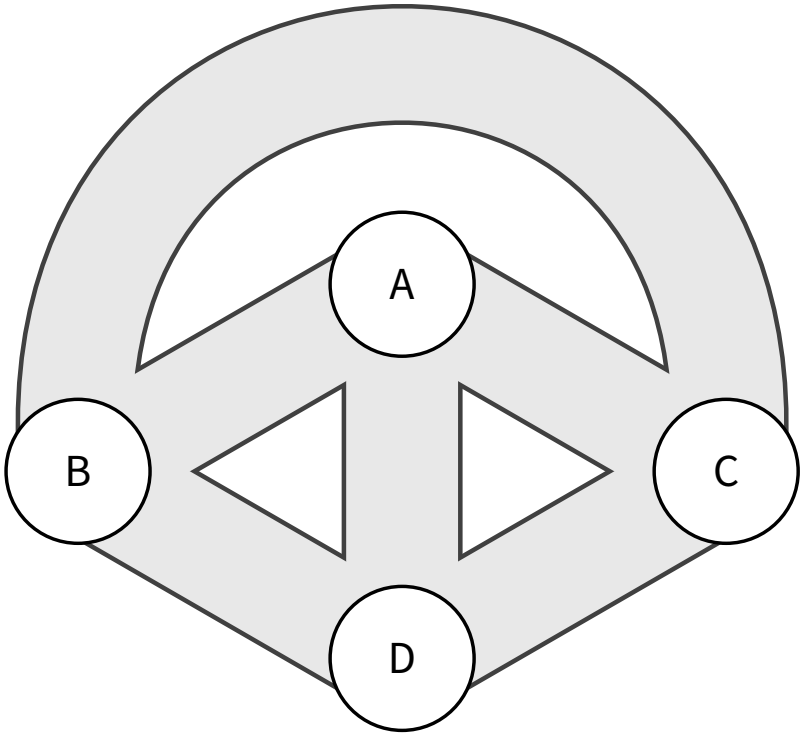


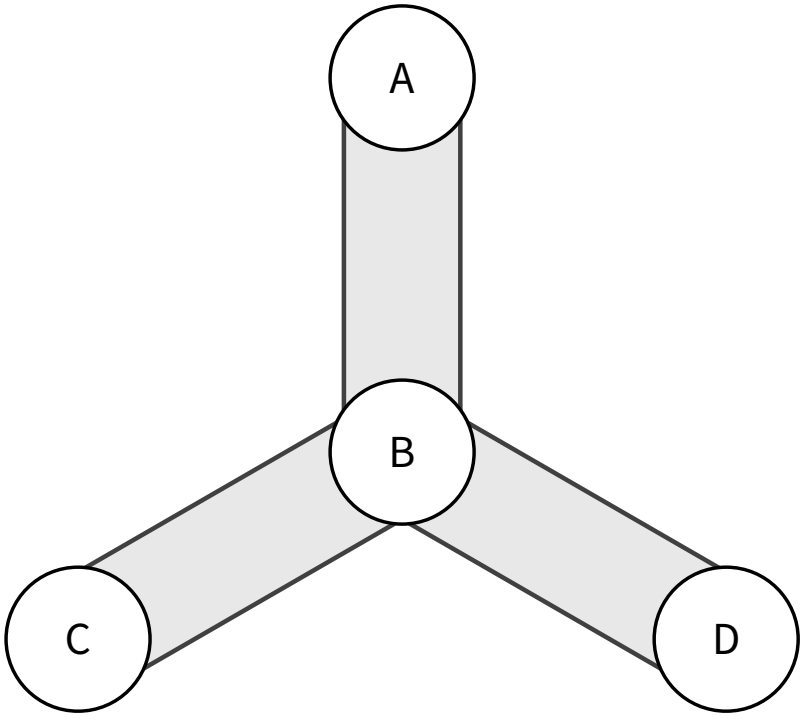
Problem 8: Lee walked around the outside of this town: A B C D E F G H I A. Find a walk from A back to A that crosses every bridge exactly once, and in which Lee's nine bridges still come in the order Lee crossed them.

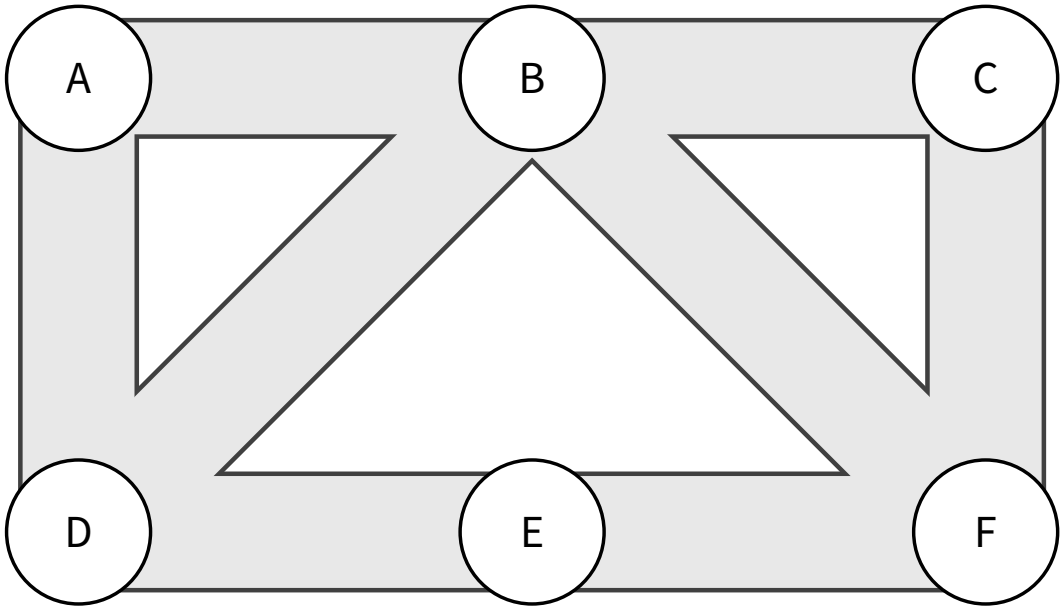


Problem 9: Explain why a town in which every island has an even number of bridges always has a walk that crosses every bridge exactly once and ends where it started. Then explain why a town with exactly two islands that have an odd number of bridges always has a walk that crosses every bridge exactly once.

Problem 10: A truck must cross every bridge of a town at least once. Before it starts, you may put extra counters on bridges, and a bridge with two counters is crossed two times. For each town, find the fewest extra counters for a route that ends where it started, and the fewest for a route that may end anywhere.







Problem 11: In the first town of Problem 10, explain why no route that ends where it started can use fewer extra counters than your answer.